What Can Machine Learning Predict in NASDAQ Markets?

Price, Direction and Volatility Forecasting Across Models and Timeframes

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AI, Machine Learning and Big Data for Finance and Management · Waseda University
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Executive Summary

Volatility is clearly predictable one candle ahead; returns and candle direction are barely so

Problem

Market moves are hard to forecast, but not every characteristic of the market need be equally hard. Which one — the next return, the next candle's direction or the next candle's realized volatility — can machine learning actually predict?

Innovation

One pipeline, one data set and one test period for three different targets: the same models are compared across tasks and timeframes, so the answer to "what is predictable" does not depend on a different set-up per task.

Usefulness

Volatility forecasts serve risk limits, position sizing, stop distances and volatility-sensitive decisions (e.g. whether to hold long or short straddles). Direction and return forecasts at this horizon are too weak to support trading after costs.

Method

Nasdaq-100 futures (NQ), 1-minute data 2006–2025, rebuilt into 5 candle lengths (5 min to 1 day). 15 models, from naive benchmarks to deep networks and Mistral's Timestral, trained on 2006–2017, tuned and selected on 2017–2021, tested once on 2021–2025.

Main findings

Volatility: ML explains +40% (5 min) and +52% (1 hour) of the error left by persistence; RV UP/DOWN accuracy 72.2% vs 51.1% (5 min); 74.8% vs 47.7% (1 hour). Return: R^2 −0.04% vs the zero forecast. Direction: 50.59% vs 49.90% majority (DNN 256-128-64-32, 5 min), but Long & Short −$515.9k after $2.50 costs. Mistral's Timestral (zero-shot) beat no benchmark except volatility at 5 min.

Conclusion

Realized volatility is by far the most predictable characteristic. The evidence for direction and returns is weak and fragile. The test period was already seen in earlier project phases, so all results are exploratory.

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Data & Methodology

One instrument, five candle lengths, three targets, one chronological split

NASDAQ historical data
NQ 1-min, 2006–2025

→ **Feature engineering**
lags, momentum, volatility, time of day

→ **Models**
benchmarks · ML · DNN · Timestral

→ **Forecasts**
for candle t+1

→ **Out-of-sample evaluation**
test 2021-07 → 2025-05

Three targets, all for the next candle (t+1), decided at the close of t

Target	Definition	Evaluated with
1 Next return	r = log(C(t+1)/C(t)), regression	MAE, RMSE, out-of-sample R^2 vs zero forecast
2 Next-candle direction	UP if C(t+1) > O(t+1); doji kept, counts as a miss	accuracy, balanced accuracy, F1 vs majority class
3 Future realized volatility	RV = Σ (1-min log returns)² inside candle t+1	log-RV error (R^2) vs persistence; RV UP if RV(t+1) > RV(t)

- **Instrument:** E-mini Nasdaq-100 futures (NQ, CME), back-adjusted continuous series, $20 per point. No QQQ, MNQ, cash index or VIX.
- **Leakage control:** features use candles ≤ t only; scaling fitted on training data; labels never cross a session or a split boundary (purged).
- **Selection:** hyper-parameters, early stopping and the five configurations chosen on validation only; test scored once (volatility score revised once, slide 7).

Candles per timeframe (rows with a valid next candle)

Candle	Train 2006-01 → 2017-08	Validation 2017-08 → 2021-07	Test 2021-07 → 2025-05
5 min	769,058	269,167	273,340
15 min	263,691	88,336	90,189
30 min	132,606	44,513	44,395
1 hour	64,813	21,642	21,507
1 day	1,991	822	642

Features (23): past returns and momentum, rolling volatility, range, body, wicks, RSI, moving-average gaps, relative volume, time of day, and log realized variance of the last 1, 5 and 22 candles plus the same time slot on the 5 previous sessions. Timestral sees only the last 512 closes. 1-minute candles excluded: their realized variance would be a single squared return.

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All Models Tested

15 models on 5 timeframes: 150 model–timeframe–task configurations, all run on the same data

Family	Model	Method	Tasks
Baselines	**Zero return / training mean**	naive forecast	R
	Majority class	always the most frequent training class	D
	Persistence	RV(t+1) = RV(t)	V
	Rolling mean (22 candles)	average past variance	V
Statistical	**HAR-OLS**	linear regression on past log RV (1, 5, 22 candles, same time slot)	V
	Ridge regression	L2-penalised linear, α chosen on validation	R · V
	Logistic regression	L2-penalised, C chosen on validation	D
Machine learning	**Random forest**	200 trees, depth 8	R · D · V
	XGBoost	gradient boosting, depth 4, early stopping	R · D · V
Deep learning	**DNN 64-32-16 (original)**	feed-forward, ReLU, AdamW, early stopping	R · D · V
	DNN 64-32		R · D · V
	DNN 128-64-32		R · D · V
	DNN 256-128-64-32		R · D · V
Mistral AI	**Timestral-5M (pretrained, zero-shot)**	forecasts quantiles of the next close from 512 past closes; median → return & direction, q10–q90 spread → volatility (calibrated on validation)	R · D · V

Notes

- **R** return · **D** direction · **V** volatility
- **Mistral's role:** Timestral is a time-series foundation model (local weights, no API, no training here). It is not a language model and it does not read news. Scored on 20,000 random validation and 20,000 test origins per timeframe (all daily ones).
- **Timestral outcome (test):** return R^2 −2.5 / −2.2 / −1.5 / −1.6 / −1.8% and direction −0.6 / +0.0 / −1.8 / −2.3 / −1.9 pp vs majority (5 min → 1 day); volatility better than persistence only at 5 min. Not among the five.
- **Considered, not run:** LSTM (the feature networks already reach their early-stopping optimum within a few epochs), larger Timestral sizes (28M–209M need new downloads).
- **Not used:** Mistral chat / LLM API — no such integration exists in the repository.

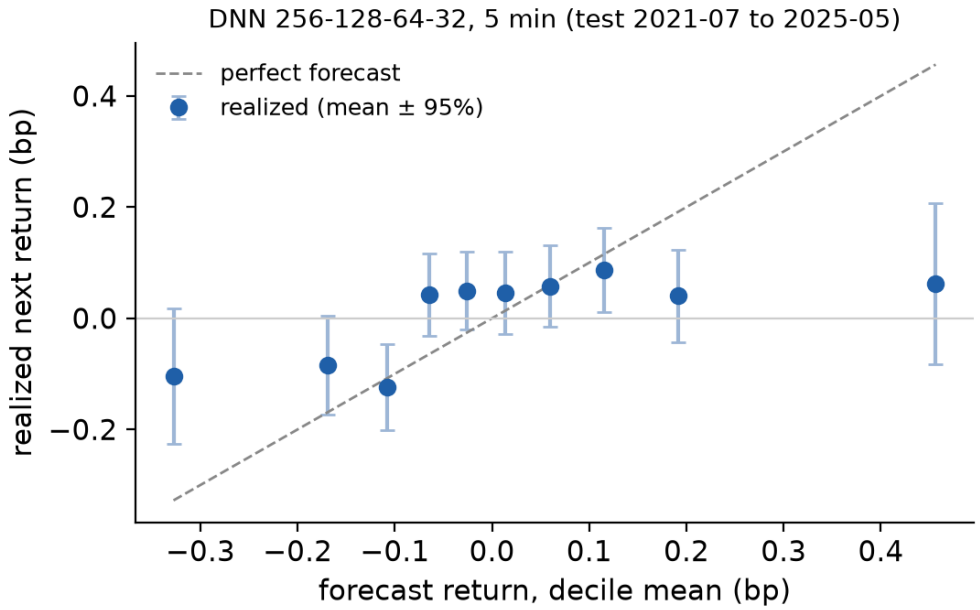
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Price / Return Forecasting

Can ML beat the zero-return forecast? Out of sample, no: its error equals that of forecasting zero

#	Model, timeframe	MAE (bp)	RMSE (bp)	RMSE zero (bp)	R² vs zero	95% interval	Sign right
1	**DNN 256-128-64-32, 5 min**	4.49	7.71	7.71	**−0.043%**	[−0.113%, +0.015%]	51.3%

- **Result:** DNN 256-128-64-32, 5 min: R² −0.043%, interval not distinguishable from zero. A positive R² would mean a smaller squared error than always forecasting zero.
- **Predicted vs realized (right):** realized returns per forecast decile. The forecasts span less than ±0.5 bp, while a typical candle moves about 7.7 bp (RMSE): the signal is buried in noise.
- **Economic reading:** no usable return signal: at these horizons NQ returns behave close to a random walk, as efficient-market reasoning suggests. Every return model and Timestral were tested on all five timeframes (inventory, slide 4).



bp = basis point = 0.01%. Intervals: 95% moving-block bootstrap over 20-session blocks. Test 2021-07 → 2025-05 (exploratory: seen in earlier project phases).

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Direction Classification & Trading

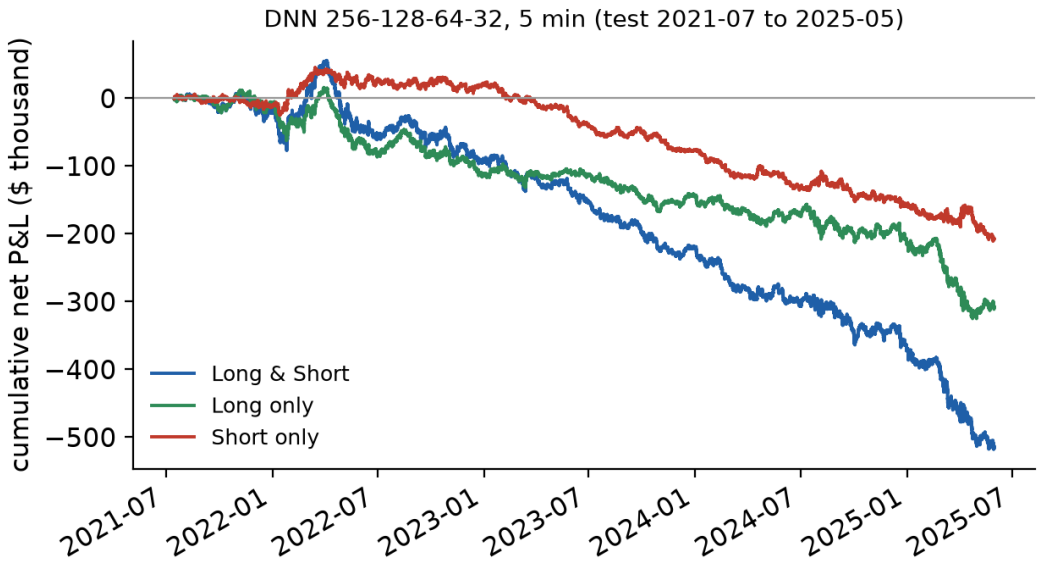
Slightly better than the majority class, but the edge per trade is smaller than the commission

#	Model, timeframe	Accuracy	Balanced acc.	Majority	Gain	95% interval	F1 up
2	**DNN 256-128-64-32, 5 min**	**50.59%**	51.37%	49.90%	+0.69 pp	[+0.40, +0.98]	0.559
3	**DNN 64-32, 15 min**	**50.99%**	50.83%	50.74%	+0.25 pp	[−0.07, +0.57]	0.622

Confusion matrix, DNN 256-128-64-32, 5 min (non-doji test candles): predicted UP & was UP 82,434 · UP but DOWN 76,180 · DOWN & DOWN 55,837 · DOWN but UP 53,956; dojis 4,933.

Trading rule, DNN 256-128-64-32, 5 min: UP → long, DOWN → short, next open → close

Strategy	Trades	Net P&L	Win rate	Profit factor	Max DD	Net / trade	Net at $5
Long & Short	273,340	**−$515.9k**	50.6%	0.98	−$573.7k	−$1.89	−$1.20M
Long only	161,587	**−$307.7k**	51.0%	0.98	−$341.0k	−$1.90	−$711.6k
Short only	111,753	**−$208.3k**	50.0%	0.98	−$256.1k	−$1.86	−$487.7k



- **Reading:** accuracy gain +0.69 pp (interval above zero). Before costs Long & Short earns +$167.4k on 273,340 trades, $0.61 per trade; the $2.50 round trip turns it into −$515.9k.
- **Accuracy ≠ profit:** with moves of size W, a hit rate p earns (2p − 1)·W − cost per trade; at p ≈ 51% and a 5-minute W of a few points the edge is cents, the cost is dollars. Fills are idealised.

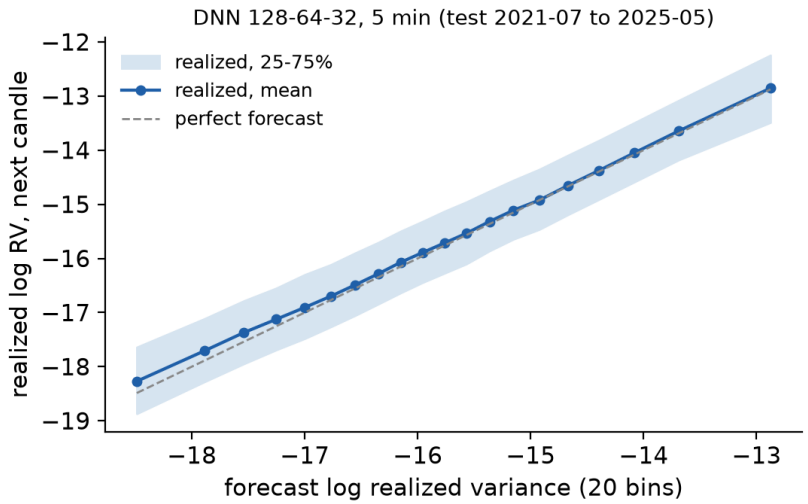
Trading: idealised fills at the next open, zero latency, 100% fills, zero slippage, $2.50 per round trip, 1 NQ = $20/point

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Volatility Forecasting

Next-candle realized volatility is strongly predictable, well beyond persistence

#	Model, timeframe	RMSE log RV	Persist.	HAR-OLS	R^2 vs persist.	QLIKE model / HAR	RV dir. acc.	Majority	Bal. acc.
4	**DNN 128-64-32, 5 min**	**0.901**	1.163	0.933	**+40.0% [+39.3, +40.6]**	0.455 / 0.477	**72.2%**	51.1%	72.0%
5	**XGBoost, 1 hour**	**0.572**	0.825	0.666	**+52.0% [+49.5, +54.8]**	0.264 / 0.275	**74.8%**	47.7%	75.1%



- **Method:** realized variance of the next candle from its own 1-minute squared log returns (r^2 is a variance proxy; volatility = √variance). Models forecast log RV; R^2 = share of persistence's squared error removed. QLIKE (variance level, lower is better) shown as a check.
- **Why it works:** volatility clusters and mean-reverts, and intraday it follows a strong time-of-day pattern. The linear HAR-OLS model already removes 36% / 35%; the ML models add 4 / 17 points.
- **Direction of volatility:** RV UP if RV(t+1) > RV(t). DNN 128-64-32, 5 min: 72.2% vs 51.1% majority; XGBoost, 1 hour: 74.8% vs 47.7% majority.

Options relevance (qualitative): rising realized volatility favours long straddles / strangles, falling volatility favours short ones. But option prices embed IMPLIED volatility: a correct realized-volatility forecast is not a profitable options trade unless it beats what the market already prices. No option prices or P&L were computed.

Selection score changed after a first report: QLIKE vs persistence was dominated on validation by a few limit-locked candles (March 2020) with ~zero variance, so log-RV squared error is used. Interval: 95% block bootstrap.

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Which NASDAQ Market Characteristic Is Most Predictable?

Five configurations selected on validation, scored once on the 2021–2025 test

ID	Task	Model	Timeframe	Key metric (test)	Baseline	Improvement [95% interval]
1	**Return**	DNN 256-128-64-32	5 min	RMSE 7.71 bp	Zero return: RMSE 7.71 bp	**R^2 -0.04% [-0.11, +0.02]**
2	**Direction**	DNN 256-128-64-32	5 min	Accuracy 50.59%	Majority: 49.90%	**+0.69 pp [+0.40, +0.98]**
3	**Direction**	DNN 64-32	15 min	Accuracy 50.99%	Majority: 50.74%	**+0.25 pp [-0.07, +0.57]**
4	**Volatility**	DNN 128-64-32	5 min	log-RV RMSE 0.901; RV UP/DOWN 72.2%	Persistence 1.163; majority 51.1%	**R^2 +40.0% [+39.3, +40.6]; direction +21.1 pp**
5	**Volatility**	XGBoost	1 hour	log-RV RMSE 0.572; RV UP/DOWN 74.8%	Persistence 0.825; majority 47.7%	**R^2 +52.0% [+49.5, +54.8]; direction +27.1 pp**

Strongest return approach
DNN 256-128-64-32, 5 min: R^2 −0.043%, not distinguishable from zero. Practically a random walk.

Strongest direction approach
DNN 256-128-64-32, 5 min: +0.69 pp over the majority class, interval above zero; Long & Short −$515.9k after $2.50 per round trip.

Strongest volatility approach
DNN 128-64-32, 5 min: R^2 +40% vs persistence, RV direction 72.2% vs 51.1%; XGBoost, 1 hour: R^2 +52% vs persistence, RV direction 74.8% vs 47.7%.

Verdict: realized volatility is the most predictable characteristic. Its gain over the benchmark is large on both selected timeframes, with intervals far from zero. The return forecast does not beat zero, and the direction gain is under one percentage point and loses money after costs. Metrics differ by task, so the ranking rests on benchmark-relative gains and their uncertainty, not raw numbers. Test 2021–2025 is exploratory.

Originality one pipeline for three targets, five timeframes, classical, deep and foundation models. **Usefulness** volatility forecasts for risk and sizing; no trading edge claimed. **Feasibility** runs on a laptop CPU, free local models, reproducible from the repository.

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